THE STANDARD AFFINE COVER OF PROJECTIVE SPACE OVER Z

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Topology of Proj. algebra-lemma-topology-proj (cross-chapter, named). D_+(T_i) cover Proj(S) because the T_i generate S_+.

Lemma 0.1. *Projective n-space over $\mathbf{Z}$ is covered by $n+1$ standard opens*

$$\mathbf{P}^n_{\mathbf{Z}} = \bigcup\nolimits_{i=0,\ldots,n} D_+(T_i)$$

where each $D_+(T_i)$ is isomorphic to $\mathbf{A}^n_{\mathbf{Z}}$ affine n-space over $\mathbf{Z}$.

Proof. This is true because $\mathbf{Z}[T_0, \ldots, T_n]_+ = (T_0, \ldots, T_n)$ and since

$$\operatorname{Spec}\left(\mathbf{Z}\left[\frac{T_0}{T_i}, \ldots, \frac{T_n}{T_i}\right]\right) \cong \mathbf{A}^n_{\mathbf{Z}}$$

in an obvious way. □

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